Problem Statement No.25

AI-Powered Signal Quality Analyzer

The Challenge Reliable telecommunication systems depend on signal quality parameters such as Signal-toNoise Ratio (SNR), Bit Error Rate (BER), and latency. Engineers must analyze these metrics to detect signal degradation and communication failures. Manual analysis of signal measurement data can be complex and time-consuming, especially in large communication networks. The challenge is to develop an intelligent system that can analyze signal performance data and provide actionable insights for improving communication quality.

The Objective Build an AI-powered Signal Quality Analyzer that evaluates signal performance and provides recommendations to improve network reliability. The solution should:

1. Process Signal Data :- Analyze signal measurement datasets containing parameters such as SNR, BER, and latency.

2. Detect Signal Issues :- Identify anomalies, interference patterns, and signal degradation.

3. Generate Intelligent Insights :- Use AI to explain the causes of signal problems and recommend improvements.

4. Predict Potential Failures :- Forecast possible signal quality issues using historical data analysis. Tech Stack • IBM Bob + IBM Models